Unmitigable Seismic and Geologic Hazards at the Proposed Crescent Gardens Cemetery Site, Meadowbrook, Riverside County, California: A Comprehensive Technical and Legal Analysis Demonstrating Significant, Persistent Risks Under CEQA and State Seismic Safety Laws

20 March 2026

Abstract

The proposed Crescent Gardens Cemetery (Conditional Use Permit No. 230002, is planned for a 46.4-acre portion of an 84.99-acre site in the Meadowbrook community of unincorporated Riverside County, California, near Perris Valley, Lake Elsinore, and Canyon Lake.

The location lies within or immediately adjacent to designated **Earthquake Zones of Required Investigation** under the **Alquist-Priolo Earthquake Fault Zoning Act** (California Public Resources Code §§ 2621 et seq.) and liquefaction hazard zones per the **Seismic Hazards Mapping Act** (Public Resources Code §§ 2690 et seq.).

This comprehensive report synthesizes all available technical data, probabilistic hazard assessments, historical seismicity, and project-specific documentation to demonstrate that the site is subject to significant, persistent, and unmitigable geologic and seismic hazards. These include surface fault rupture risk, strong ground shaking, liquefaction susceptibility in alluvial soils, cumulative effects of frequent small earthquakes, and heightened vulnerability from proposed imported backfill of unknown composition.

The Draft Mitigated Negative Declaration (MND) circulated by Riverside County Planning Department in July 2025 fails to adequately address or reduce these impacts to less-than-significant levels as required by the **California Environmental Quality Act (CEQA)** (Public Resources Code §§ 21000 et seq.).

Substantial evidence supports the conclusion that approval of the conditional use permit would violate state mandates to protect public health, safety, welfare, environmental quality, and the dignity of burial sites (Government Code § 65850 et seq.; CEQA Guidelines § 15126.4). Denial of Conditional Use Permit No. 230002 or preparation of a full Environmental Impact Report is legally warranted and necessary.

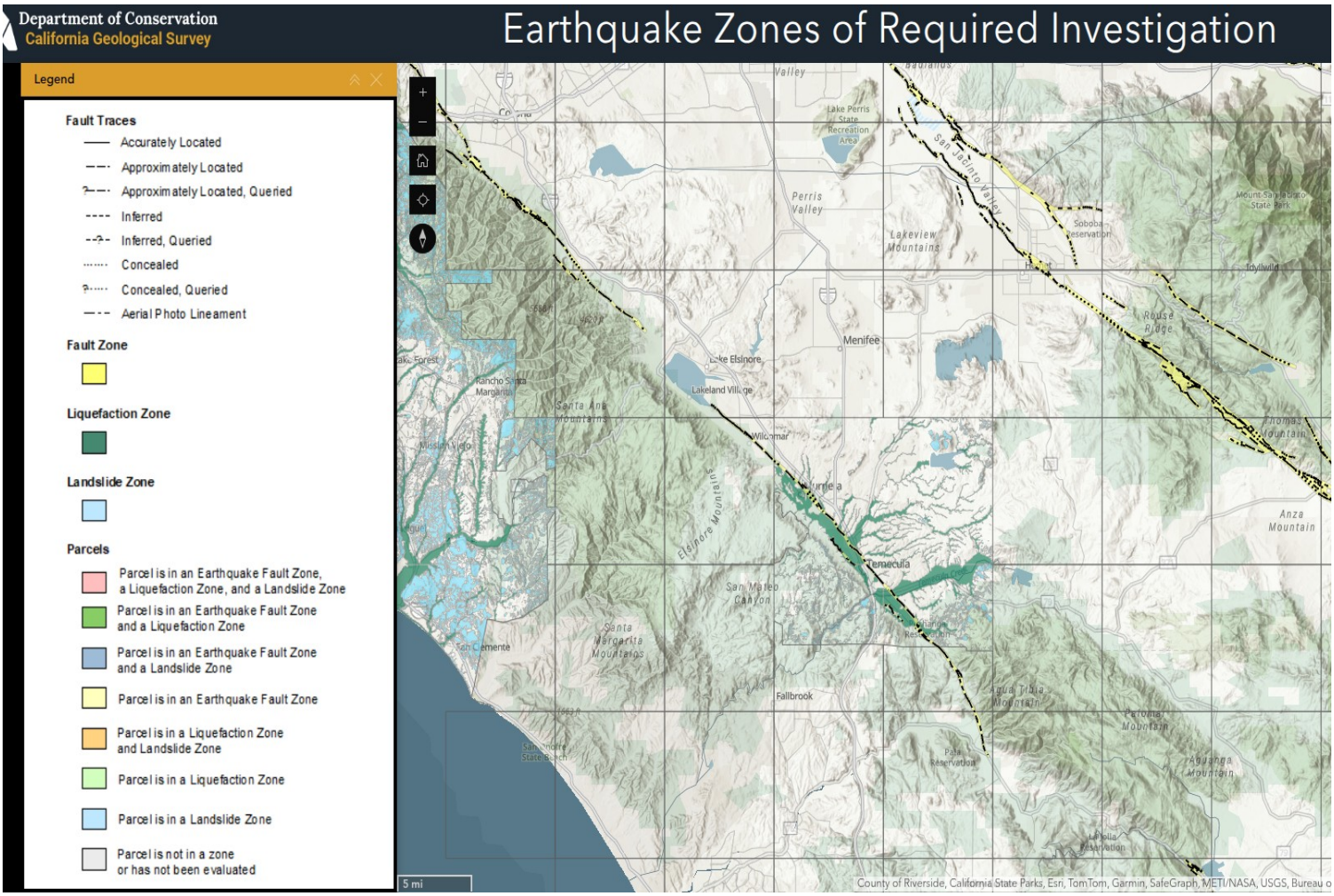
1. Introduction and Current Project Status

The Crescent Gardens Cemetery project, proposed by the Muslim Mortuary and Cemetery Committee, envisions a phased 15-year development of approximately 20,000 vaulted burial plots, a mortuary, chapel, ceremonial cleansing facility, caretaker residence, equipment shed, internal roads, septic system, water quality basins, and 145 parking spaces. Riverside County Planning Department, acting as CEQA lead agency, issued a Draft MND in July 2025 with public review concluding August 28, 2025, asserting that geologic, seismic, and related impacts would be mitigated to less-than-significant levels through referenced geotechnical measures.

As of March 20, 2026, the project remains under review with no final approval or denial recorded. Recent activity includes Canyon Lake City Council discussions on March 11, 2026, ongoing opposition petitions highlighting groundwater contamination, traffic, and neighborhood character concerns, and a counter-petition in support. The absence of a final decision preserves the opportunity—and legal necessity—for rigorous scrutiny of unaddressed hazards.

2. Geologic and Seismic Setting

The site is situated in alluvial valley deposits along the Elsinore Fault Zone, a major right-lateral strike-slip fault with an approximate slip rate of 4 mm/year and the capacity to generate earthquakes up to magnitude 7.5 or greater in multi-segment ruptures. California Geological Survey hazard maps clearly depict the area in yellow (Earthquake Fault Zone) and green (liquefaction zone) designations, indicating parcels requiring mandatory site-specific geotechnical investigations for surface fault rupture, liquefaction, and seismically induced landslides. Project geotechnical data note perched groundwater at 13–15 feet below surface and deeper main groundwater at approximately 147 feet, conditions that heighten liquefaction potential in loose, saturated granular soils.



3. Probabilistic and Historical Seismic Hazards
The Third Uniform California Earthquake Rupture Forecast (UCERF3) (USGS Open-File Report 2013-1165; California Geological Survey Special Report 228) assigns the Elsinore Fault Zone a 30-year probability of approximately 3–4% for an M ≥ 6.7 event. Moderate earthquakes (M ≥ 5.2) are more frequent, with site-relevant probabilities for moderate local shaking (peak ground acceleration 0.15g–0.30g) estimated at roughly 1–5% over 30 years from proximal fault sources, supplemented by background seismicity. No magnitude 5.2 earthquake has been recorded directly at the site in instrumental history; the most recent relevant event on the fault zone was April 14, 2025 (M5.2 near Julian, approximately 40–70 miles southeast and felt in the region).

4. Impacts from Moderate Earthquakes (M5.2 Scenario)
A magnitude 5.2 earthquake could generate moderate to strong shaking (Modified Mercalli Intensity VI–VII) sufficient to exceed liquefaction initiation thresholds in the site's susceptible alluvial soils. Resulting ground failure would include differential settlement, lateral spreading, and buoyancy forces capable of causing concrete burial vaults to float, tilt, crack, or displace by inches to feet. Breaks in proposed water, sewer, and septic lines would inject additional fluids into the soil, increasing saturation, pore water pressure, and ground deformation in a destructive feedback loop that further compromises vault integrity, grave stability, and site infrastructure.

5. Cumulative Effects of Frequent Small Earthquakes and Microseismicity
The region experiences frequent small earthquakes and microseismicity (typically M1.0 to M2.5) on a daily to weekly basis due to ongoing tectonic strain along the Elsinore Fault and surrounding structures. While individual events produce minimal perceptible shaking, repeated low-level cyclic loading over decades induces progressive fatigue in soil structure and buried elements. In liquefaction-prone alluvial deposits, this manifests as gradual pore pressure accumulation, subtle densification or loosening of soil layers, and incremental differential settlement. Over the operational life of a cemetery, such chronic effects can accumulate to inches of uneven settlement, loosening pipe joints, initiating micro-cracks in concrete vaults, accelerating corrosion in utilities, and preconditioning the site for more severe failure during moderate or large events. These long-term, insidious processes are not adequately evaluated in the Draft MND or referenced geotechnical report, despite their relevance under the **Seismic Hazards Mapping Act** and CEQA's requirement to disclose all significant impacts (Public Resources Code §§ 2690 et seq.; CEQA Guidelines § 15126).

6. Exacerbation by Imported Backfill Soils of Unknown Composition
Construction phases contemplate the importation of substantial quantities of fill material (e.g., approximately 30,500 cubic yards in Phase 2). Backfill soils of unknown composition introduce heterogeneous layers with unpredictable grain-size distribution, density, plasticity, and liquefaction susceptibility. Such untested materials can replicate or worsen the loose, saturated conditions that trigger liquefaction at lower shaking intensities, amplify non-uniform settlement, and increase the likelihood of vault displacement and utility rupture. This practice contravenes California Building Code Chapter 18 requirements for engineered fill and undermines the assumptions of the project’s

geotechnical mitigation strategy, rendering proposed earthen buffers and vault designs insufficient to achieve less-than-significant impacts.

7. Inadequacy of Proposed Mitigation and Legal Non-Compliance

The Draft MND relies on a geotechnical report claiming that earthen buffers, vault construction, and other measures will reduce geologic and seismic impacts to less-than-significant levels. However, liquefaction and seismically induced settlement frequently produce non-uniform, unpredictable deformation that exceeds the reliability of standard mitigation techniques (California Geological Survey Special Publication 117A). CEQA imposes a substantive duty on lead agencies to avoid or substantially lessen significant environmental effects where feasible (Public Resources Code §§ 21000 et seq.; CEQA Guidelines § 15126.4). The substantial evidence presented, California Geological Survey hazard mapping, UCERF3 probabilistic assessments, historical liquefaction case studies, pipeline fatigue data, and the added risks from unknown backfill and chronic microseismicity, demonstrates that residual hazards cannot be reliably mitigated to negligible levels.

8. Conclusions and Legal Recommendations

The proposed Crescent Gardens Cemetery site is burdened by unmitigable seismic and geologic hazards that threaten public safety, the long-term integrity and dignity of burial sites, groundwater quality, and surrounding environmental resources. Approval of Conditional Use Permit No. 230002 would violate Riverside County's fundamental obligation to protect public health, safety, and welfare (Government Code § 65850 et seq.) and contravene CEQA's mandate to prevent significant unmitigated impacts.

The County must deny the conditional use permit or, at minimum, prepare a full Environmental Impact Report to transparently disclose and analyze these persistent risks. All interested parties are urged to submit formal comments and evidence to the Riverside County Planning Commission and Board of Supervisors, citing the technical and legal grounds outlined herein, to ensure the project is not approved and constructed in this seismically vulnerable location.